BMC64-NG — Differences from BMX and the original BMC64

The differences, one by one

Up to date as of 11 October 2026. For BMC64 this means version 5.1.1 (the guide and the sources we have), with the news from 5.2 where they are mentioned. For BMX this means version 2026.08.19, the last one in its changelog.

The three systems

- **BMC64** is Randy Rossi's project: a Commodore emulator that runs on the Raspberry Pi by itself, with no operating system (Circle), with VICE 3.3. It runs on Raspberry Pi Zero, Zero 2 W, 2 and 3. The Pi 4 and later are not supported.
- **BMC64-NG** starts from BMC64, but uses VICE 3.10 and runs on Raspberry Pi 4, 400, 5, 500 and 500+. The older Raspberry Pi models are gone.
- **BMX** is another fork of BMC64, also with VICE 3.10, for Raspberry Pi 4 and 5 (and their variants). It dropped the Raspberry Pi models up to the 3.

BMC64-NG shares with BMC64 the menu, the `machines.txt` file format, the REU images saved and loaded from the menu, and the writing of disks back to the card (Prefs > Flush disk writes). It shares only two things with BMX: VICE 3.10 and the CRT shader.

BMC64-NG compared with the original BMC64, Randy's

Machines and Raspberry Pi models

- **Raspberry Pi**: BMC64-NG runs on Pi 4, 400, 5, 500 and 500+. BMC64 runs on Pi Zero, Zero 2 W, 2 and 3, and not on the Pi 4 or later.
- **Emulator**: BMC64-NG has VICE 3.10, BMC64 has the 2018 VICE 3.3.
- **Cycle-exact C64**: in BMC64-NG the C64 is cycle exact (X64SC), with the lighter Std version selectable in the Machine > Switch menu. BMC64 only has the normal C64.
- **SuperCPU**: BMC64-NG has the C64 with a 20 MHz SuperCPU. BMC64 does not.
- **C64 models**: from the Model... menu the C64 can be a C64C, an old C64, an SX-64 or a C64 GS (the cartridge-only console). BMC64 does not offer them: its VICE knows them, but the menu does not offer them.
- **Plus4Emu**: in BMC64-NG it runs on the Raspberry Pi 4, 400, 5, 500 and 500+. In BMC64 it is Pi 3 only.

Restart, power off, keys

- **Power off**: BMC64 is switched off by pulling the power. BMC64-NG also powers off and restarts from the Reset menu and from the keyboard (**ALT + ESC**, **ALT + SHIFT + ESC**, with confirmation), with **Fn + F10** on the Pi 400 and with the power button of the Pi 500 and 500+.
- **Shortcuts**: BMC64-NG has about thirty shortcuts with ALT (disks, tape, cartridges, REU, drive model, volume, SID player, scanlines, warp, status bar). BMC64 has the C= and CTRL combinations with F1, F3, F5, F7 and the buttons to which a function can be assigned, which BMC64-NG kept.
- **Keyboards**: BMC64-NG has American, Italian, British and German layouts (Keyboard > Layout menu), also for typing in the menu fields. BMC64 has the American one, the positional one and the Maxi one.
- **RESTORE on two keys**: in the Positional mappings of the C64, C128, VIC-20 and SuperCPU RESTORE is **PrtScn** (the Pi 400 and 500 keyboards) and also **Page Up** (a real C64 keyboard on a Keyrah V2). BMC64 has only Page Up.

Video

- **Main screen**: Video > Main HDMI Output chooses which HDMI port carries the picture, and the audio with it. BMC64 does not: it has only one screen.
- **C128 on two monitors**: Video > Active Display set to Two HDMI sends the VIC-II to one monitor and the VDC to the other. BMC64 does not: VIC-II and VDC go to a single output, with Active Display and PIP.
- **Composite output**: BMC64-NG has **PAL Composite 576p@50Hz** and **NTSC Composite 480p@60Hz** in every machine's Switch menu, only on the Pi 4 B (3.5 mm jack, 4-pole AV cable), with the borders kept apart from the HDMI ones. BMC64 has the same two composite modes for its Raspberry Pi models.
- **VGA**: BMC64-NG has 1024x768 VGA modes at 50 and 60 Hz for every machine (HDMI with a VGA adapter). BMC64 takes another route (DPI with a VGA666, or 1920x1080 with `raster_skip`).
- **VGA and HDMI monitors together**: Video > VGA Monitor On tells which HDMI port the VGA monitor is on, and Switch sends the screen to the VGA with the VGA entries and to the other monitor with the others, with a single restart. BMC64 does not: it has only one screen.
- **CRT shader**: BMC64-NG has BMX's shader, with geometry, scanlines, phosphor mask, bloom, reflections and the `.crt` presets. BMC64 has davej's CRT filter.
- **Video mode confirmation**: after a machine or mode change there are 15 seconds to confirm with a key, otherwise it goes back to what it was (Safety timer (15s) at the bottom of the Switch menu). BMC64 does not: it asks for confirmation only for resets, and to go back it has CTRL + F7 held for 5 seconds.

Sound

- **SID engines**: besides ReSid there is **ReSIDfp** (Sound > SID Engine), with its own settings, the 6581 composer profiles (Composer Profile) and the background noise (Background Noise). BMC64 does not: its SID Engine has only Fast and ReSid.
- **Resampling**: SID Resampling is set at the factory to Resampling, the most faithful. In BMC64 on the Pi 3 you choose between Fast, Interpolation and Fast Resampling, and on the Pi Zero and 2 there is only Fast.

- **reSID filters**: in the Sound menu the reSID Filter Settings folder (Passband %, Gain %, Filter Bias) for the 6581 and the 8580. In BMC64's guide Passband and Gain must stay at the factory values.
- **Stereo or mono** (Sound > Channels) with more than one SID.
- **SID Card on PET and Plus/4**, off at the factory. BMC64 does not have it: the second SID exists only on the C64 and the C128.
- **Autostart on the PET**, with `.prg` files only. BMC64 does not have it: there is no Autostart on the PET.
- **.SID file player**: ALT + S opens the `/sids` folder; it also plays files with two, three, four and up to eight SIDs (PSID v4E; with eight-SID tunes, such as The Tuneful Eight, 24 bars and reSID to start with; S switches between reSID and reSIDfp live), with VU meters, song time, 10-second jumps (back at once, from a snapshot of the machine kept in memory), voices one by one and the side of each SID (F1-F4, F1-F8 with eight-SID tunes). BMC64 does not have it.
- **SIDs on more cores**: with 2 or 3 SIDs, and in the player up to 8, the ones after the first are computed by core 2 and, when the screen leaves it free, by core 3. BMC64 does not: its extra cores only compute the filter tables at startup, then stay idle.
- **USB DAC**: Sound > Audio out chooses between Auto, Jack, HDMI and USB DAC from the menu. The DAC can be plugged and unplugged while the machine is on, and 48 kHz-only DACs work too. BMC64 does not: the audio output (HDMI, jack or automatic) is chosen only in `machines.txt`, with no USB DAC.
- **Volume**: the on-screen bar, **ALT + "+"**, **ALT + "−"**, **ALT + M**.

Clock

- **CPU and GPU clock from the menu** (CPU Clock and GPU Clock): the CPU -20%, -10%, stock, +10%, +20%, the GPU stock, +10%, +20%, on every Pi from the Pi 4 on; the GPU MHz at the top of the menu and on the status bar, next to the CPU ones. In BMC64 the clock is changed by hand in `config.txt` (the `arm_freq` and `over_voltage` lines, commented out in the sample file).

Disks, tapes, files

- **Real Commodore drive**: a 1541, 1571 or 1581 on an XUM1541 or ZoomFloppy adapter (ALT + 7). BMC64 does not have it.
- **USB floppy drives from a PC** read as FLP1 and FLP2. BMC64 does not have them.
- **Search in the file list**: you type the first letters, with wildcards (the question mark and the asterisk), and if nothing is found it searches inside the name. BMC64 only jumps to the first letter typed.
- **Long names in the menu**: a name that does not fit in the line scrolls when the cursor is on it, so you can read it all (it stops for a moment, scrolls, then starts again). BMC64 cuts it.
- **.REU images**: **ALT + 0** loads a file from the `/REU` folder, **ALT + SHIFT + 0** detaches it.
- **EasyFlash cartridges save** (Cartridge > Save EasyFlash Now, and by themselves when the cartridge is detached). BMC64 has Save EasyFlash Now, but does not save by itself when the cartridge is detached.
- **Drive without a disk in Plus4Emu**: ALT + 2 and ALT + 3 set the 1541 or the 1551 and open the list of `.D64` files.

- **.NIB images** (the low-level copies made with nibtools, protected disks included): attached and started like the other disks, converted to `.g64` with the nibconv code of nibtools. BMC64 does not have them.

Network

- **Network on all machines**, also on the Pi 5 and the Pi 500, from the Network menu. BMC64 has the network only on C64 and C128.
- **WiFi from the menu**: WiFi Settings > WiFi SSID searches for networks and lists them with band, channel and signal; WiFi Country Code is typed in; the password is typed in clear text in the box and after the reboot it is shown as asterisks.
- **Fixed address**: IP Address Mode set to DHCP or Static (Static IP, Netmask, Gateway, DNS). The Raspberry Pi introduces itself to the router as BMC64-NG. BMC64 does not: only the automatic address (DHCP).
- **Daylight saving time**: Daylight Saving (Off, On, Auto (EU)) under Timezone. BMC64 takes the time from the network and has a time zone in minutes in the settings file (`timezone_offset_minutes`); with no daylight saving time.
- **Date and time**: at the top of the main menu, taken from the network with the time zone and daylight saving of the Network menu; the files seen through SD Card Sharing get the same time. BMC64 takes the time from the network but does not show it.
- **SD Card Sharing...**: the microSD seen from the PC (`\\BMC64-NG-400` , with the name of the model, or `\\address` , user bmc64, password bmc64, changeable with Sharing Password). BMC64 does not have it.
- **NAS**: the NAS entry (NAS IP Address, NAS User Name, NAS Password, SMB Protocol, NAS Path, Test NAS Connection) opens a NAS the way the MiSTer does; NAS Path is the share and the folder the browser starts from, one per machine, empty at the factory like the other fields; the password is typed in clear text and after the reboot it is shown as asterisks. BMC64 does not have it.
- **The modem like the Ultimate II+** (C64, C128, Plus/4 and VIC-20): Modem (SwiftLink) off at the factory, Modem Address `DE00` at the factory. It also receives calls, on port 3000 (RING, **ATA**, **ATS0=1**). BMC64's modem only dials out; its factory address is `DE00` in 5.1.1 and `D700` from 5.2. The modem on the Plus/4 (the built-in ACIA, `$FD00`) and on the VIC-20 (the user port) is in BMC64 too, from 5.2.6. On the C64 and the C128 the line is on DSR, as on the Ultimate II+; in BMC64 it is on DCD.

Controls

- **The user port joystick adapter**: in BMC64-NG the Starbyte can be chosen (in BMC64 it is in the list but the menu never reaches it) and after a snapshot the menu finds the right type again (BMC64 only looks at the first one).
- **Spinner as a paddle** (Joyports > USB Spinner (Paddle), with Spinner Sensitivity) on C64, C128 and VIC-20. BMC64 does not: as a paddle it uses only the analog axes of gamepads.
- **The menu by blocks with ALT**: in the menu **ALT** + ↑ and ↓ jump up and down by blocks like **PgUp** and **PgDn** (held down, they keep going), with any keyboard: the 500+ too, where FN does not reach the computer. In BMC64 **ALT** does nothing in the menu, and ↑ and ↓ move one row.
- **Tape controls with no tape**: with no tape mounted the recorder controls do nothing and **NO TAPE** appears at the bottom.

- **Status bar**: the real lights of each drive model, the track (half-tracks too), the tape counter, and temperature and frequency on the second line (**ALT + SHIFT + O**).

When something goes wrong

- **The watchdog**: if the emulator stays still for ten seconds a large black box appears, in English (BMC64-NG HAS STOPPED), with the address for reporting the defect on GitHub.
- **Files on the card**: `BMC64-DIAG.TXT`, `PASSI.TXT` (and `PASSI-PRIMA.TXT`), `BLOCCO.TXT` (and `BLOCCO-PRIMA.TXT`), `USB-LOG.TXT`, `NET-LOG.TXT`. BMC64 has its own log, switched on from the main menu and from the configuration file.

BMC64-NG compared with BMX

Raspberry Pi and machines

- **Raspberry Pi**: BMC64-NG is tested on Pi 4, 400, 5, 500 and 500+. BMX prepares the card also for the CM4 and, on the Pi 4, can build its own old 32-bit version.
- **Plus/4**: BMC64-NG has two engines, VICE and Plus4Emu. BMX only VICE.
- **Machines**: BMC64-NG has C64 (cycle exact and Std), SuperCPU, C128, VIC-20, Plus/4 and PET; BMX has the same, with the cycle-exact C64 and the SuperCPU marked as experimental.
- **Changing machine**: BMC64-NG from the Machine > Switch menu with the profiles of `machines.txt`, with 15 seconds to confirm; BMX from the Machine menu with Apply & Reboot, with the profiles of `machines.ini` (it does not read `machines.txt`), with no timed confirmation.

Restart, power off, keys

- **Restart and power off**: BMC64-NG also from the keyboard (**ALT + ESC**, **ALT + SHIFT + ESC**, **Fn + F10** on the Pi 400, the power button on the Pi 500 and 500+). BMX only from the System menu.
- **Shortcuts**: all those with ALT belong to BMC64-NG. BMX only has F12 and the eight C= and CTRL + function key combinations, which BMC64-NG has too.
- **Keyboards**: BMC64-NG American, Italian, British and German; BMX American and German.
- **RESTORE on two keys**: BMC64-NG **PrtScn** (Pi 400 and 500) and **Page Up** (Keyrah V2); BMX only Page Up.

Clock

- **CPU clock**: BMC64-NG from the CPU Clock menu, from -20% to +20% of the stock clock of every Pi from the Pi 4 on, so also lower. BMX only upwards (Pi 4 from 1500 to 2400 MHz, Pi 5 from 2400 to 3200 MHz, in steps of 25 MHz).
- **GPU clock**: BMC64-NG from the main menu (GPU Clock), Core and V3D together, stock, +10% or +20%, with the GPU MHz at the top of the menu and on the status bar. BMX has separate Core Clock and V3D Clock in the Expert folder, also only upwards (Pi 4 up to 800 MHz, Pi 5 up to 1200).

Video

- **Main screen**: in BMC64-NG Video > Main HDMI Output chooses HDMI 0 or 1 for all machines. In BMX the firmware decides.
- **C128 on two monitors**: BMC64-NG yes (Two HDMI, with swapping). BMX shows the two screens on one monitor only.
- **VGA**: BMC64-NG has 1024x768 profiles at 50 and 60 Hz. BMX does not.
- **VGA and HDMI monitors together**: in BMC64-NG Video > VGA Monitor On moves the screen between the HDMI monitor and the VGA one, on the other port, with the Switch entry alone. BMX does not: the firmware picks the port.
- **Composite output**: BMC64-NG has **PAL Composite 576p@50Hz** and **NTSC Composite 480p@60Hz** in the Pi 4 B Switch menu, with the borders kept apart from the HDMI ones, and you go back to HDMI with an HDMI entry of Switch or with C= + F7 held for 5 seconds. BMX has two composite modes (576p and 480p) marked as experimental.
- **Video mode confirmation**: 15 seconds to confirm, then it goes back to what it was (Safety timer (15s) removes it). BMX does not have it.
- **Video modes**: BMC64-NG has 720p at 50 and 60 Hz (1080p for the PET) and VGA 1024x768. BMX has 480p and 576p over HDMI, 720p, 1080p and its own exact-frequency modes.
- **VIC-20 NTSC** with its American KERNAL: BMC64-NG yes. BMX has NTSC timing, but does not choose an NTSC KERNAL.

Sound

- **Outputs**: BMC64-NG HDMI, USB DAC and the 3.5 mm jack of the Pi 4. BMX HDMI and USB DAC.
- **SID Card on PET and Plus/4**: BMC64-NG only ($8F00 or $E900 on the PET, $FD40 or $FE80 on the Plus/4).
- **.SID file player** (ALT + S) up to eight SIDs: BMC64-NG only. BMX does not open `.SID` files.
- **ReSIDfp** as a third SID engine (Sound > SID Engine), with Background Noise and the 6581 composer profiles: BMC64-NG. BMX does not: in its menu the engine has only FastSID and reSID.
- **reSID filters**: BMC64-NG has the reSID Filter Settings folder, valid for all SIDs. BMX shows the values on screen and adjusts the second SID on its own.
- **SIDs on more cores**: BMC64-NG with core 2 and, when the screen leaves it free, with core 3, with 2 or 3 SIDs and in the player up to 8; BMX only the second SID on core 2, with two SIDs.
- **Stereo or mono** (Sound > Channels): BMC64-NG. In BMX stereo goes with the second SID.

Files, disks and cartridges

- **Real Commodore drive** (XUM1541, ZoomFloppy, ALT + 7): BMC64-NG only.
- **USB floppy drives** (FLP1 and FLP2) and the CMD FD-2000 and FD-4000 drives: BMC64-NG.
- **Search in the file list** with wildcards (the question mark and the asterisk): BMC64-NG.
- **Long names in the menu**: in BMC64-NG they scroll under the cursor. BMX cuts them.
- **.REU images**: BMC64-NG with **ALT + 0** from the `/REU` folder and from the menu. BMX only from the REU menu.
- **.prg programs**: in BMC64-NG they start even without a drive. BMX uses a temporary disk and needs a drive.

- **.SID files**: BMC64-NG only.
- **.NIB images**: BMC64-NG only (converted to `.g64` with the nibconv code of nibtools).
- **Saving disks**: Prefs > Flush disk writes in BMC64-NG, and EasyFlash cartridges save.

Network

- **Network**: BMC64-NG has Wi-Fi or Ethernet on all machines, also on the Pi 5 and 500; WiFi from the menu with a list of networks; automatic or fixed address; daylight saving time; date and time taken from the network, at the top of the menu. BMX has Wi-Fi or Ethernet with a fixed address and network search; its menu shows no time.
- **SD Card Sharing...**: the microSD seen from the PC over SMB (`\\BMC64-NG-400`, with the name of the model), with the Raspberry Pi in the Windows Network folder. BMX does not have it.
- **NAS like the MiSTer**: BMC64-NG only. BMX does not have it.
- **Modem like the Ultimate II+** (SwiftLink, dials and receives): BMC64-NG, on the C64, C128, Plus/4 and VIC-20. BMX has a Hayes modem with a phone book and sounds, RS-232 over the network, UP9600 and SwiftLink.
- **Web commands and update from the Internet**: BMX only. In BMC64-NG there is no web control, by choice.

Controls

- **The user port joystick adapter** (Joyports > Userport Joystick Adapter): in BMC64-NG the types you can choose are the ones VICE 3.10 has for the machine, Synergy and WOJ on the Plus/4. BMX keeps the VICE 3.3 list: on the Plus/4 only types that 3.10 refuses are left, and the adapter never turns on; on the C64 the Hummer is left, which in 3.10 belongs to the C64DTV only.
- **Spinner as a paddle** (Joyports > USB Spinner (Paddle)): BMC64-NG on C64, C128 and VIC-20. BMX has nothing about the spinner (searched in its sources).
- **The menu by blocks with ALT**: in the menu **ALT** + ↑ and ↓ jump up and down by blocks like **PgUp** and **PgDn** (held down, they keep going), with any keyboard, the 500+ too. In BMX **ALT** does nothing in the menu, and ↑ and ↓ move one row.
- **Tape controls with no tape**: BMC64-NG shows **NO TAPE** and does nothing. In BMX the tape controls reach VICE unchecked.
- **Tape sound** (Tape sound emulation) does not wipe out the SID music in BMC64-NG; in the VICE file of BMX the defect is still there (our comparison of 19 September).
- **VICE fixes after 3.10**: BMC64-NG has all of BMX's and, since 4 October 2026, the other fidelity fixes of the development VICE: the CIA serial register, the extra RTS read, the PET's original 80-column editor, the $FFF0 register of the PET 8096/8296, the C128 VDC lock-up with R0=0, the C128's custom C64 kernal (no longer replaced by the stock one), the C128 $01 bits, two TED registers, the VIC-20 joystick and keyboard column 7, the POT glitch only when a connected POT changes, the drive ROM changed from the menu reaching the running drive at once, the virtual drive's T-W commands, SFX Sound Expander, DigiMAX, tapecart and the PET HRE board. Since 10 October 2026 also the keyboard clear that forgets the key queue (r46275 of 28 September).

When something goes wrong

- **BMC64-NG** writes `BMC64-DIAG.TXT`, `PASSI.TXT`, `BLOCCO.TXT`, `USB-LOG.TXT` and `NET-LOG.TXT` to the card, and has the **watchdog** with the black BMC64-NG HAS STOPPED box. BMX has an on-screen diagnostics panel, with no files and no watchdog.

What they have and we do not (or not yet)

BMX

- **Heavy overclocking**: CPU up to 2400 MHz on the Pi 4 and 3200 MHz on the Pi 5, **Voltage Offset**, **Temperature Limit** and, in the Expert folder, **Core Clock** and **V3D Clock** (GPU clocks). BMC64-NG's CPU Clock stops at +20%. Core and V3D are also adjusted by BMC64-NG, with its GPU Clock (stock, +10%, +20%).
- **480p and 576p over HDMI** (720x480 at 60 Hz and 720x576 at 50 Hz) as video modes for all machines.
- **Card with two partitions** (SYS: and USER:), a disk chosen to be mounted at every power-on and a utility disk (with ccgms) in drive 9.
- **Menu**: the five Quick Access slots, the menu used with the mouse, the enlargeable menu, the pending system changes shown before the restart (Pending system changes).
- **Keyboard map editor** and on-screen monitors for keyboard, mouse and GPIO.
- **Hayes modem** with a phone book and sounds, RS-232 over the network, UP9600 and Userport.
- **Web commands** (off at the factory) and **update from GitHub**.
- **On-screen diagnostics panel** with frames, memory and core load.
- **Palettes loaded from the card** (`palettes`) and **contents of disks and tapes** shown in the file list.
- **Declared 8BitDo controllers** (Ultimate 2, Ultimate C, V1 and V2 dongles, keyboards): not tested in BMC64-NG.

Randy's BMC64

- **Raspberry Pi Zero, Zero 2 W, 2 and 3**: BMC64-NG runs on none of the three.
- **Web UI and updater** (BMC64 5.2): a web interface on the local network with an SD card file browser, a BASIC program editor and an updater using `bmc64-update.zip`. BMC64-NG does not have them: the owner ruled out the Web UI.
- **DPI output** (VGA666, RGB on the comb) with its modes in `machines.txt`: in BMC64-NG the code is there, but the modes in `machines.txt` are not.
- **Choice of the card partition** (`disk_partition=` in `cmdline.txt`): BMC64-NG has it too, but without the fix from BMC64 5.2.0 (#369).
- **Large CMD HD and IDE64 images** (over 32 MB) read from the card as needed: in BMC64 since 5.2; in BMC64-NG, as of 16 September, still to do.